OPIPE-01: OpenPipeline as a Multi-Scope Platform

> **Series:** OPIPE — OpenPipeline Beyond Logs | **Notebook:** 1 of 7 | **Created:** March 2026 | **Last Updated:** 10/26/2026

Beyond Logs: Processing Spans, Metrics, and Events at Ingestion

OpenPipeline is often introduced as a log processing framework — and logs are indeed the most common entry point. But OpenPipeline operates on many **configuration scopes** — logs, spans, metrics, events, business events, security events and more — each with its own pipelines, routing rules, and extraction capabilities. This notebook builds intuition for how the platform works across all scopes and why pipeline design matters.

> **Companion Series**
> - **OPLOGS** — Deep-dive into log processing: **OPLOGS-01: OpenPipeline Fundamentals**
> - **OPMIG** — Migrating from classic logs? Start with **OPMIG-01: OpenPipeline Migration Guide: Part 1**
> - **SPANS** — Querying and analyzing distributed traces: **SPANS-01: Fundamentals**

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Grail enabled
Permissions	`storage:logs:read`, `storage:spans:read`, `storage:metrics:read`, `storage:events:read`, `storage:bizevents:read`
Recommended	Familiarity with **OPLOGS-01** (log pipeline basics)

OneAgent 1.333+: Primary Fields/Tags as Routing Keys

From **OneAgent 1.333**, OneAgent can enrich telemetry at the source with standardized **primary fields** and customer-defined **primary tags** as top-level attributes — an OpenPipeline-relevant change, because those attributes are present before routing. The agent version is what matters here, not the tenant version: verify the OneAgent version on the hosts concerned (details and configuration in OneAgent Attribute Enrichment below).

Primary fields (Semantic Dictionary-defined):

Field	Purpose
`dt.security_context`	ABAC scope for IAM policies; also drives bucket-routing decisions
`dt.cost.costcenter`	Cost allocation tag; routes spend to org units
`dt.cost.product`	Product-line attribution for cost rollups

Primary tags (customer-defined): set during OneAgent install via `oneagentctl --set-host-tag="primary_tags.<key>=<value>"` — the `primary_tags.` prefix must be written explicitly (it is never added automatically); e.g., `primary_tags.team`, `primary_tags.env`,

`primary_tags.app`, `primary_tags.data_classification`. Land as `primary_tags.<key>` on every signal, up to 20 primary tags per host or process — beyond 20, *"OneAgent doesn't emit any warning and does not enrich data with the excess tags."*

Pipeline implication for cross-scope design:

1. **Routing rules** can dispatch on primary fields directly without parse processors. A route's matching condition is DQL, and a route sends records to a **pipeline**, not to a bucket:

Route matcher (DQL)	Target pipeline
`dt.cost.costcenter == "cc-1234"`	`finance-logs`
`matchesValue(dt.security_context, "*pci*")`	`pci-audit` (whose Bucket assignment stage picks the 365-day bucket)

The bucket is chosen later, by the target pipeline's **Bucket assignment** stage. Write substring tests as `matchesValue(field, "*text*")`, not `contains()`: OpenPipeline matchers accept only a subset of DQL functions, and a condition written with `contains()` or `in()` is rejected with the verifier error `The function contains() isn't enabled.` `matchesValue` is case-insensitive and also matches any element when `dt.security_context` holds an array value (see the array rule in §6).

2. **Cross-scope consistency:** because primary fields/tags appear on logs, spans, metrics, AND business events, queries that join across scopes (OPIPE-06 cross-scope design patterns) can correlate without scope-specific lookup tables.

3. **Non-OneAgent sources** (raw syslog, third-party log shippers, OTLP-via-collector) still need OpenPipeline `enrichment` processors to surface the same fields. Document this split in your standard.

> **Update (June 2026) — primary tags are now formally documented.** Dynatrace published a dedicated Primary tags (DT docs) page that canonicalizes the model. Two points matter for pipeline design: (1) **OpenPipeline itself is a documented primary-tag source** — `primary_tags.*` values can be *"derived or transformed from any incoming field at ingest-processing time"*, which closes the non-OneAgent-sources gap in point 3 above: instead of generic enrichment processors producing ad-hoc fields, derive proper `primary_tags.<key>` values so downstream routing and bucket assignment treat agent and agentless data identically. (2) The other documented sources are OneAgent, Kubernetes annotations (`metadata.dynatrace.com/primary_tags.<key>`), cloud-provider tags, OpenTelemetry resource attributes, and host or process metadata. Primary tags are "available before

> data enters the processing pipeline," which is exactly what makes them usable in routing conditions like the examples above.

SaaS 1.337 added recommended-field suggestions to extraction processors. Per the release notes, extraction processors *"now supply a recommended set of fields to be extracted, which helps avoid misconfiguration and exposing sensitive data"*, and the recommendations cover permission- and cost-relevant fields, Smartscape identifiers, and Grail primary tags.

> **Sources:** Primary Grail fields and tags enrichment through OneAgent (DT docs), SaaS 1.337 release notes (DT docs).

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1. The OpenPipeline Scopes

OpenPipeline processes data through independent **configuration scopes** — logs, spans, metrics, generic events, Davis events and problems, SDLC events, business events, security events, system events (limited support), Smartscape events (limited support), user events, and user sessions. Each scope has its own set of pipelines, routing rules, processors, and storage targets. They share the same architecture but operate on different data types. This series covers the six you will configure most:

Scope	Data Object	Primary Use Case	Key Fields
Logs	`logs`	Application and infrastructure log records	`content`, `loglevel`, `log.source`
Spans	`spans`	Distributed trace spans from services	`span.kind`, `trace.id`, `duration`
Metrics	`timeseries`	Time-series measurements from agents and extensions	`metric.key`, `dt.entity.*`
Events	`events`	Lifecycle and configuration change events	`event.kind`, `event.type`
Business Events	`bizevents`	Business transactions and user actions	`event.type`, `event.provider`

Scope	Data Object	Primary Use Case	Key Fields
Security Events	`security.events`	Vulnerability, detection, and compliance findings	`event.kind`, `event.type`

> **Sources:** OpenPipeline data flow (DT docs) — configuration scope list.

What This Means in Practice

Each scope is configured **independently** in the Dynatrace UI under **OpenPipeline** settings. A processing rule you create for logs has no effect on spans. A routing rule for business events does not apply to metrics. This independence is a feature — it means you can evolve each scope's pipeline at its own pace without risk to other data types.

Discovering Your Active Scopes

The following queries show what data is flowing through each scope in your environment.

```
// Log pipelines: volume by pipeline and bucket
fetch logs, from:-1h
| summarize log_count = count(), by:{dt.openpipeline.pipelines,
dt.system.bucket}
| sort log_count desc
```

```
// Span pipelines: volume by pipeline and bucket
fetch spans, from:-1h
| summarize span_count = count(), by:{dt.openpipeline.pipelines,
dt.system.bucket}
| sort span_count desc
```

```
// Business event pipelines: volume by provider and type
fetch bizevents, from:-1h
| summarize event_count = count(), by:{event.provider, event.type}
| sort event_count desc
```

2. Shared Architecture Across Scopes

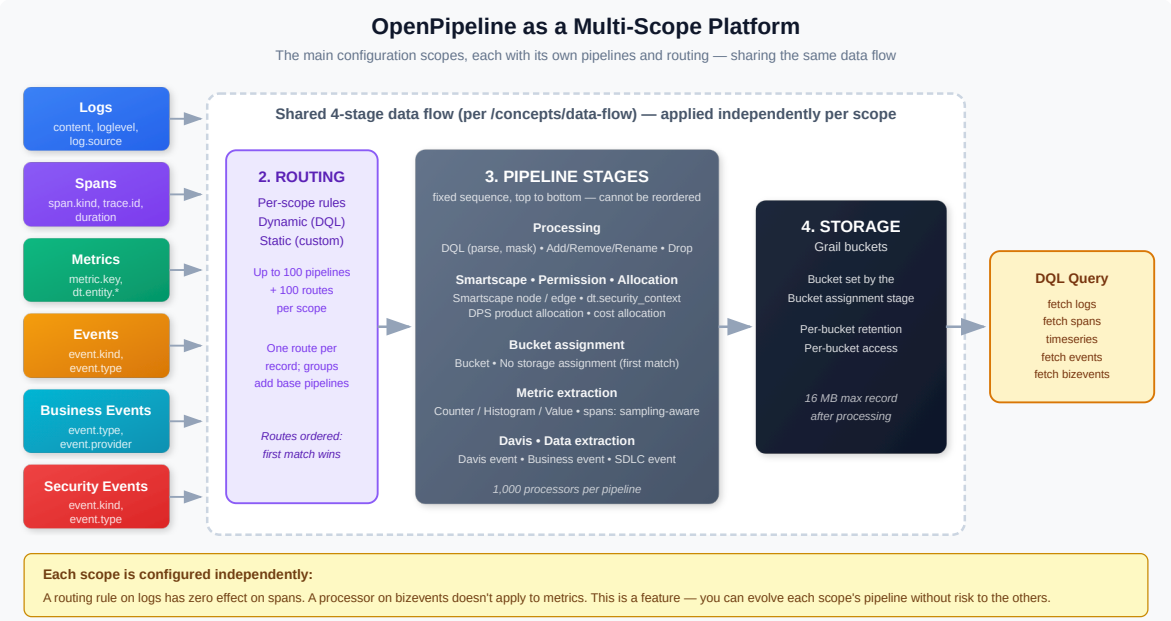
Sprint 1.345 (August 2026): Temporary Fields (`dt.temp.*`)

A field whose name is prefixed `dt.temp.` (with the dot) lives for the duration of pipeline processing and is **discarded before the record is written to Grail**. That closes a long-standing gap in the extraction pattern this series teaches: previously, an intermediate value needed for metric, event, or bizevent extraction had to be written onto the record — where it persisted, consumed storage, and changed the shape of the stored data for every downstream consumer.

Property	Behaviour
Lifetime	Available to subsequent processors in the same pipeline, from the processor that defines it; dropped at storage
Persistence	Never written to Grail — invisible to `fetch`
Cost	Not persisted, so it adds nothing to stored volume or retention
Enablement	None — the `dt.temp.` prefix alone marks the field temporary; a field named `dt.temp` on its own is stored normally
Scope support	Every scope with a Processing stage; not applicable to Smartscape upsert events

Use it for intermediate calculations feeding extraction. Do **not** use it for enrichment you intend to query later — the field will not be there. SaaS 1.345 released 08/11/2026 with a **staged tenant rollout**; verify availability in your tenant before relying on the prefix, and note that a `dt.temp.` field written on a tenant that does not yet support the semantics would simply persist as an ordinary attribute.

> **Sources:** Processing in OpenPipeline (DT docs) — *"Prefix a field name with dt.temp. to create a temporary field."*; `dt.temp` used as a field name itself *"is not a temporary field and is stored normally"*; *"Temporary fields are available to the subsequent processors within the same pipeline, starting from the processor that defines them."*; *"Temporary fields aren't applicable to Smartscape upsert events."*



Every scope follows the same **data flow** (Ingest → Routing → Processing → Storage), with optional pre-processing on custom sources. Inside a pipeline, processing is not one step: it runs as a **fixed sequence of stages**, of which the stage named *Processing* is only the first. The flow is identical in concept across scopes — only the data types and available processors differ.

Stage	Purpose	Logs Example	Spans Example
1. Ingest	Receive records from a built-in / ready-made / custom source	OneAgent log ingest	OTLP span ingest
2. Routing	Direct data to the right pipeline (dynamic DQL match or static for custom sources)	Match by `log.source` or `k8s.namespace.name`	Match by `span.kind` or `service.name`

Stage	Purpose	Logs Example	Spans Example
3. Pipeline	A fixed sequence of stages: **Processing** (DQL, add / remove / rename fields, drop record, GeoIP and inline lookup) → **Smartscape node** → **Smartscape edge** → **Permission** (security context) → **Product allocation** → **Cost allocation** → **Bucket assignment** → **Metric extraction** → **Davis** → **Data extraction**	Mask credit card numbers, drop debug, parse JSON (Processing); extract error-rate metrics (Metric extraction)	Mask PII, drop health-check spans (Processing); sampling-aware RED metrics (Metric extraction)
4. Storage	Persist to the Grail bucket chosen in the Bucket assignment stage (or skip with No storage assignment)	Send to `app_logs` bucket	Send to `trace_data` bucket

> **Stage order is fixed.** Per *Processing in OpenPipeline*, *"The sequence of stages is fixed for all pipelines and cannot be modified."* Masking, parsing, and dropping happen in the first stage (Processing); Metric extraction, Davis, and Data extraction run **after** Bucket assignment. Within a stage, processors run in the order you list them, and each stage executes either all matching processors or only the first match.
>
> **Sources:** Processing in OpenPipeline (DT docs) — stage table, in execution order; OpenPipeline limits (DT docs) — per-scope, per-pipeline and record-size limits in the diagram.

Sprint 1.346 (August 2026): `dt.bindplane.*` Is a Reserved Namespace

Verbatim: *"The `dt.bindplane.*` namespace is now reserved exclusively for Bindplane. Dynatrace uses it to populate fields such as `dt.bindplane.project`, `dt.bindplane.fleet`, and `dt.bindplane.configuration`"* — to keep those fields populated consistently.

The practical rule for pipeline authors is the same one that governs every `dt.*` namespace: **do not write into it from a processor.** If a processing rule of yours adds or overwrites a `dt.bindplane.*` field, move it to a namespace you own before the reservation

takes effect, or the platform-populated value and yours will contend. This is worth a grep of your existing pipelines rather than an assumption — the namespace was not reserved before, so nothing stopped a rule from using it.

SaaS 1.346 released 08/25/2026 with a **staged tenant rollout** from the same date; verify against your own tenant.

The Key Insight

If you understand how to build a log pipeline (route to a pipeline → mask, drop, and transform in Processing → set permissions and cost allocation → assign a bucket → extract metrics and events), you already understand how to build a span pipeline or an event pipeline. The concepts transfer directly — only the field names and processor options change.

Processors Available Per Scope

Not every processor is available in every scope. The following table shows key differences (refresh against Processing in OpenPipeline (DT docs) for the current list):

Processor	Logs	Spans	Metrics	Events	Bizevents
DPL parsing	Yes	Yes	—	Yes	Yes
Field enrichment (Add / Remove / Rename / DQL)	Yes	Yes	Yes	Yes	Yes
Metric extraction (Counter / Value / Histogram Preview)	Yes	Yes (sampling-aware variants)	—	Yes	Yes
Smartscape node / edge	Yes	Yes	Yes	Yes	Yes
Event extraction (Business / SDLC event in Data extraction; Davis event in Davis)	Yes	Yes	—	Yes	Yes
Masking (DQL processor, Processing stage)	Yes	Yes	Yes	Yes	Yes
Inline lookup (Processing stage)	Yes	Yes	Yes	Yes	Yes
GeoIP lookup (Processing stage, Early Access)	Yes	Yes	Yes	Yes	Yes

Processor	Logs	Spans	Metrics	Events	Bizevents
Drop record	Yes	Yes	Yes	Yes	Yes
Bucket assignment / No storage assignment	Yes	Yes	—	Yes	Yes
Set dt.security_context	Yes	Yes	Yes	Yes	Yes
DPS Cost Allocation (Cost Center / Product)	Yes	Yes	Yes	—	—

OneAgent Attribute Enrichment (OneAgent 1.333+)

> **Requires:** OneAgent version **1.333** or later

OneAgent can enrich **all telemetry signals** (metrics, spans, logs, events, entities) with custom metadata at the source — before data reaches the Dynatrace platform. This is more efficient than server-side tagging (auto-tags) because enrichment happens on the host and propagates to all Smartscape nodes.

Primary Fields (standardized from Semantic Dictionary):
- `dt.security_context` — data governance and access control
- `dt.cost.costcenter` — cost allocation
- `dt.cost.product` — product attribution

Primary Tags (custom key-value pairs):
- `primary_tags.environment` — environment identification (production, staging, etc.)
- `primary_tags.team` — team ownership
- `primary_tags.business_unit` — organizational unit

Configuration:

```
# Set during OneAgent installation
Dynatrace-OneAgent-Linux.sh --set-host-
tag="primary_tags.environment=production" --set-host-
tag="dt.security_context=confidential"

# Set on existing agents via oneagentctl
oneagentctl --set-host-tag="primary_tags.environment=production"
oneagentctl --set-host-tag="dt.cost.costcenter=12345"

# Per-process via environment variable (overrides host-level)
DT_TAGS="primary_tags.team=platform primary_tags.environment=production"
```

Benefits over auto-tagging:

Aspect	Auto-Tags (Server-Side)	Attribute Enrichment (Agent-Side)
When applied	After data arrives at platform	At the source, before transmission
Scope	Entity tags only	All signals: metrics, spans, logs, events, entities
Grail integration	Limited	Full — feeds OpenPipeline routing, bucket assignment, permissions
Cost allocation	Not supported	`dt.cost.costcenter`, `dt.cost.product` fields
Security context	Not supported	`dt.security_context` for data governance

> **See:** Primary Grail fields and tags enrichment through OneAgent (DT docs)

3. The Default Pipeline Anti-Pattern

Every OpenPipeline scope has a built-in **Default Pipeline** — for logs and business events, only where the Classic pipeline is not available. It is a catch-all: any data that does not match a custom pipeline's routing rules flows through the default pipeline and lands in the default bucket (e.g., `default_logs`, `default_spans`).

This is convenient for getting started. It becomes a serious problem at scale — Dynatrace's own guidance is to *"limit the use of the default pipeline to monitoring unassigned incoming data"*.

> **Sources:** Processing in OpenPipeline (DT docs) — *"It's not available for log and business event configuration scopes where the Classic pipeline is available."*

Why "Everything in Default" Fails

Problem	Impact
Query performance	Every DQL query against the default bucket scans ALL data — application logs mixed with infrastructure noise, health checks mixed with business transactions. `fetch` reads at most 500 GB of uncompressed data by default (`scanLimitGBytes`), and an undifferentiated bucket reaches that sooner.
Cost	No differentiated retention. You pay to store debug-level container stdout at the same retention tier as critical security audit logs.
Security	No `dt.security_context` separation. Without dedicated pipelines, you cannot apply IAM policies to restrict who sees which data. Finance team logs are visible to infrastructure engineers and vice versa.
Blast radius	A misconfigured processing rule (bad parsing regex, aggressive drop filter) in the default pipeline affects ALL data flowing through it. One mistake breaks everything.
No metric extraction	You cannot extract targeted metrics from specific log patterns when everything is in one undifferentiated stream. Metric extraction rules become overly complex or extract noise.
Debugging	When something goes wrong, you cannot tell which data source caused the issue. Pipeline-level monitoring (`dt.openpipeline.pipelines`) shows a single pipeline name for everything.

The Telltale Signs

Run the following queries to check if your environment has this problem.

```
// Check: What percentage of logs land in the default bucket?
fetch logs, from:-1h
| summarize total = count(),
    default_count = countIf(dt.system.bucket == "default_logs")
| fieldsAdd default_pct = round(toDouble(default_count) / toDouble(total) *
100, decimals: 1)
| fieldsAdd assessment = if(default_pct &gt; 80, then: "CRITICAL: Most data in
default bucket",
    else: if(default_pct &gt; 50, then: "WARNING: Majority in default bucket",
    else: "OK: Data is distributed across buckets"))
```

```
// Check: How many distinct pipelines are processing your logs?
fetch logs, from:-1h
| summarize pipeline_count = countDistinct(dt.openpipeline.pipelines),
    bucket_count = countDistinct(dt.system.bucket),
    source_count = countDistinct(dt.openpipeline.source)
| fieldsAdd assessment = if(pipeline_count &lt;= 1, then: "CRITICAL: Only
default pipeline in use",
    else: if(pipeline_count &lt; 3, then: "WARNING: Very few pipelines
configured",
    else: "OK: Multiple pipelines configured"))
```

```
// Visualize: Volume distribution across buckets (are they balanced?)
//
// The grand total must be computed BEFORE the rows are split apart. After a
grouped
// summarize, log_count is a scalar per row — not an array — so
arraySum(log_count) is
// a type mismatch. The query still returns ok:true and rows, and every
pct_of_total
// reads null, with only a DATATYPE_MISMATCH warning to signal it. Collect the
rows
// into one array, total them there, then expand back to one row per bucket.
fetch logs, from:-1h
| summarize log_count = count(), by:{dt.system.bucket}
| summarize rows = collectArray(record(bucket = dt.system.bucket, log_count =
log_count)),
            grand_total = sum(log_count)
| expand rows
| fieldsAdd bucket = rows[bucket], log_count = rows[log_count]
| fieldsAdd pct_of_total = round(100.0 * log_count / grand_total, decimals: 1)
| fields bucket, log_count, pct_of_total
| sort log_count desc
```

```
// Same check for spans: are they all in default_spans?
fetch spans, from:-1h
| summarize total = count(),
    default_count = countIf(dt.system.bucket == "default_spans")
| fieldsAdd default_pct = round(toDouble(default_count) / toDouble(total) *
100, decimals: 1)
| fieldsAdd assessment = if(default_pct &gt; 80, then: "CRITICAL: Most spans in
default bucket",
    else: if(default_pct &gt; 50, then: "WARNING: Majority in default bucket",
    else: "OK: Spans are distributed across buckets"))
```

4. Pipeline Design Principles

The goal is to move from "everything in default" to a **purpose-driven pipeline architecture**. Here are the principles:

Principle 1: One Pipeline Per Source Type

Create separate pipelines based on the nature of the data, not the team that owns it.

Pipeline	Routing Condition	Target Bucket	Retention
`kubernetes-infra`	`k8s.namespace.name` exists AND `log.source == "Container Output"`	`k8s_infra_logs`	14 days
`application-logs`	`log.source == "Log file"` AND `k8s.namespace.name` matches app namespaces	`app_logs`	35 days
`security-audit`	`matchesValue(log.source, "*audit*")` or another source-identifying field (`AUDIT` is not a log level)	`security_audit`	365 days
`api-ingestion`	`dt.openpipeline.source == "/api/v2/logs/ingest"`	`external_logs`	35 days
Default route	Everything else (catch-all) — for logs, the Classic pipeline where it is available	`default_logs`	35 days (built-in bucket)

Principle 2: Filter Early, Extract Targeted

Each pipeline should drop noise **before** storage and extract **only** the metrics relevant to that data type:

- **kubernetes-infra pipeline**: Drop health check logs, extract pod restart counts
- **application-logs pipeline**: Drop DEBUG in production, extract error rates by service
- **security-audit pipeline**: Never drop, never sample — extract compliance event counts

Principle 3: Differentiate Retention

Different data has different value over time:

Data Type	Suggested Retention	Rationale
Debug/container stdout	7-14 days	High volume, short-term troubleshooting
Application logs	35 days	Standard operational window
Security/audit logs	365+ days	Compliance requirements (SOC 2, HIPAA)
Business events	90-365 days	Trend analysis, quarterly reporting

Principle 4: Route Ordering Matters

Pipeline routing rules are evaluated **in order — first match wins**: *"If no route matches the record, the record is routed via the Default route."* Place specific rules before general ones:

1. Security audit logs (most specific, highest value)
2. Application logs by namespace
3. Kubernetes infrastructure
4. API-ingested external logs
5. Default route (catch-all for anything unmatched)

Applying This to Spans

The same principles apply to the spans scope. `service.name` is an OpenTelemetry attribute and is absent on OneAgent spans, which carry `dt.service.name` instead — match on whichever your spans actually have (OPIPE-02 §1):

Pipeline	Routing Condition	Target Bucket
`frontend-traces`	`span.kind == "server"` AND `service.name` matches frontend services	`frontend_spans`
`backend-traces`	`span.kind == "server"` AND `service.name` matches backend services	`backend_spans`
`database-spans`	`span.kind == "client"` AND `db.system` exists	`db_spans`
Default Pipeline	Everything else	`default_spans`

```
// Inventory: What log sources exist and how much volume do they produce?
// Use this to plan your pipeline routing rules
fetch logs, from:-24h
| summarize {log_count = count(), unique_hosts =
countDistinct(dt.entity.host)},
    by:{dt.openpipeline.source, log.source}
| sort log_count desc
| limit 20
```

```
// Inventory: Kubernetes namespace volume (for pipeline routing decisions)
fetch logs, from:-24h
| filter isNotNull(k8s.namespace.name)
| summarize log_count = count(), by:{k8s.namespace.name}
| sort log_count desc
| limit 20
```

```
// Inventory: Span volume by service (for span pipeline routing decisions)
fetch spans, from:-24h
| summarize span_count = count(), by:{dt.service.name, span.kind}
| sort span_count desc
| limit 20
```

5. Matching Conditions: Conditional Logic Within a Pipeline

Sections 3 and 4 explained why you need separate pipelines and how to design them. But what happens when a single pipeline receives data from several sources that need **different processing logic** — without justifying an entirely new pipeline?

The answer is the **matching condition** every processor carries.

How Conditional Processing Works

There is no grouping construct for processors inside a stage. Each processor has its own matcher: *"it defines the target of a processor via a DQL statement and narrows down the available data to the specific set you want to process."* On the DQL processor the matching condition is a required field.

Behavior	Detail
Matching	Each processor's matching condition is a DQL statement. Only records that satisfy it are processed; the rest pass through unchanged.
Order	Processors run in the order listed: *"each processor output becomes the input for the next one."* A field one processor adds is available to the matchers of the processors after it.
All matches vs. first match	The stage decides. Processing, Smartscape, Metric extraction, Davis and Data extraction run **every** matching processor; Permission, Product allocation, Cost allocation and Bucket assignment run only the **first** match.
"Every record"	A matcher of `true` applies the processor to all records.
Matcher language	A subset of DQL — `matchesValue`, `matchesPhrase`, `isNull`, `isNotNull`, comparisons and boolean logic. `contains()` and `in()` are rejected (see the References note).

> **SaaS 1.344+:** OpenPipeline matchers support **Duration**-type record fields, with the operators `=`, `!=`, `>`, `>=`, `<`, and `<=` (rollout from 07/29/2026). On a tenant that has not yet reached 1.344, match on an equivalent **numeric** field instead (for example a millisecond-valued field carried alongside the record rather than the duration field itself).
>
> **Sources:** Processing in OpenPipeline (DT docs) — processor anatomy and the per-stage "All matches" / "First match only" column, Processing stage (DT docs) — *"Set a specific matching condition on each processor to narrow down the records it applies to."*, DQL matcher in OpenPipeline (DT docs) — *"true — the processor (DQL query) will be applied to all records"*.

Parse Once, Match on the Result

Because there is no shared group matcher, a condition used by several processors is repeated on each one. Keep it cheap and keep it in one place: parse the distinguishing value once, early in the Processing stage, into a field, and let later processors match on that field. Dynatrace's own guidance: *"Parse it once and store the extracted values in fields for downstream processors to reuse."* If the field is only needed during processing, name it `dt.temp.*` (SaaS 1.345+, §2) so it is not stored.

Example: Multi-Format Log Pipeline

A single `application-logs` pipeline receives logs from Java, Node.js and Python services. Each has a different format, but they all go to the same `app_logs` bucket with 35-day retention. One pipeline, with each processor scoped by its own matcher:

Stage	Processor	Matching condition	What it does
Processing	DQL	`matchesValue(k8s.container.name, "java-*")`	Parse log4j format, extract exception class, add `app.framework = "java"`
Processing	DQL	`matchesValue(k8s.container.name, "node-*")`	Parse JSON structured logs, extract request ID, add `app.framework = "nodejs"`
Processing	DQL	`matchesValue(k8s.container.name, "python-*")`	Parse Python logging format, extract traceback, add `app.framework = "python"`
Processing	Drop record	`loglevel == "DEBUG"`	Drop debug records from every source
Permission	Security context	`true`	Set `dt.security_context = "app-team"`
Bucket assignment	Bucket assignment	`true`	`app_logs`

Three parsing paths, one pipeline, one bucket. The alternative — three pipelines with three routes — triples the configuration surface for data whose lifecycle is identical.

Example: Span Enrichment by Service Category

In the Spans scope, a single pipeline processes server and client spans but needs different enrichment by category:

Processor matching condition	Processing
`matchesValue(http.route, "/api/*")`	Add `span.category = "api"`, extract API version from path
`isNotNull(db.system)`	Add `span.category = "database"`, normalize the statement to remove parameters
`span.kind == "producer" or span.kind == "consumer"`	Add `span.category = "messaging"`, extract queue name

Matching Conditions, Pipelines, or Pipeline Groups?

Scenario	Use
Same bucket, retention and owner; different parsing per source	**Matching conditions** on processors in one pipeline
Only the destination bucket differs	**Bucket assignment** processors with conditions in one pipeline (first match wins)
Different retention, security context, or data lifecycle	**Separate pipelines**, each with its own route
Different teams own the configuration	**Separate pipelines** — *"By adding custom pipelines per team, you can manage them via owner-based access control."*
A central team must enforce the same stages (for example masking) on many teams' pipelines	**Pipeline groups** — a composition of base pipelines that *"can restrict or mandate stages"* for its member pipelines; **OPIPE-07** covers running one
Blast-radius isolation	**Separate pipelines** — a bad processor in one pipeline cannot affect another

> **Sources:** Processing in OpenPipeline (DT docs) — custom pipelines and owner-based access control, Pipeline groups (DT docs) — compositions, base and member pipelines.

Common Mistakes

Mistake	Problem	Fix
Re-parsing `content` in every processor's matcher or definition	Slower processing; complex DQL processors can be disabled	Parse once into a field; match on the field
Overlapping matchers in an all-matches stage without intent	A record processed by several processors may get conflicting field values	Make matchers mutually exclusive, or order them so the last write is the one you want
Overlapping matchers in a first-match stage (Permission, cost/product allocation, Bucket assignment)	Only the first processor fires; later, more specific ones never run	List the most specific condition first
One pipeline per destination bucket when only the destination differs	Pipeline proliferation for data that is otherwise processed identically	Use conditional **Bucket assignment** processors (first match wins) inside one pipeline
Dozens of source-specific processors in one pipeline	As hard to reason about as separate pipelines, without the ownership boundary	Split into separate pipelines when the sources have different owners or lifecycles

6. Security Context Across Scopes

The `dt.security_context` field controls **who can see which data** through IAM policies. It is the mechanism that makes pipeline separation actionable from a governance perspective.

How Security Context Works

When data flows through an OpenPipeline, you can enrich it with a `dt.security_context` value. IAM policies then use this value to restrict query access:

```
Data Ingestion → OpenPipeline adds dt.security_context → Grail stores it → IAM policy enforces access
```

For a complete deep-dive on security context patterns, values, and IAM policy syntax, see **ORGNZ-06: Security Context**.

Security Context by Scope

Scope	How to Set `dt.security_context`	Notes
Logs	OpenPipeline **Permission** stage	Most common — set based on `k8s.namespace.name`, `log.source`, or parsed fields
Spans	OpenPipeline **Permission** stage	Set based on `service.name` or span attributes
Metrics	OpenPipeline **Permission** stage **or** extension configuration	See "Extension Metrics" below
Events	OpenPipeline **Permission** stage	Set based on `event.type` or other event fields
Business Events	OpenPipeline **Permission** stage	Set based on `event.provider` or business unit
All signals from a host	`dt.security_context` host tag (`oneagentctl --set-host-tag`)	*"used to automatically determine the security context for all logs, spans, metrics, and events that are sent from this host"*
Entities	**Settings > Topology model > Grail Security Context**, host tag, or an extraction rule for generic types	Controls access to **entities only** — data sent from an entity does not inherit the entity's security context

Extension Metrics and Security Context

> **Extensions can now carry primary Grail tags directly** (OneAgent, ActiveGate and EEC 1.343+). A monitoring configuration accepts primary tags per configuration and per endpoint; remote extensions also accept primary fields, while local extensions inherit the host's. See **FAQ-26 §5.5** for the local/remote difference.
>
> **Sources:** Enrich extensions with primary Grail fields and tags (DT docs) — *"Remote extensions run on the Extension Execution Controller (EEC) without a OneAgent host, so enrichment cannot be inherited automatically."*

Metrics ingested via **Extensions 2.0** carry a `dt.security_context` only if something sets it. A common gap is to deploy extensions for database, cloud, or infrastructure metrics and assume the security context of the monitored entity carries over. It does not: an IAM policy that filters entities *"will not filter related metrics, logs, or traces"*.

There are two approaches:

Approach 1: Set it in the extension configuration

Dynatrace documents setting `dt.security_context` *"per extension configuration"*. The extension then sends it as a dimension on every data point, for example `postgres.activity.idle,port=5432,dt.security_context="TeamA",…`. One configuration per team gives each team's metrics their own context. To restrict access to the extension's **entities** as well, add an extraction rule that maps the same dimension onto the generic entity type.

Approach 2: OpenPipeline Permission stage on the Metrics scope

When the extension configuration cannot carry it, set `dt.security_context` in the **Metrics** scope's Permission stage, matching on the metric key or on dimensions that identify the source:

Matching condition	Security context value	Use case
`matchesValue(metric.key, "postgres.*")`	`"database-team"`	A database extension's metrics
`matchesValue(metric.key, "aws.*")`	`"cloud-team"`	Cloud integration metrics

Extension metric keys are whatever the extension defines (`postgres.*`, `sql-server.*`, …) — check the real keys with `metrics | filter startsWith(metric.key, "postgres")` before writing the matcher.

> **Sources:** Grant access to entities with security context (DT docs) — *"Logs, spans, metrics, and events powered by Grail that are sent from an entity do not inherit the management zones of that entity."*, Processing in OpenPipeline (DT docs) — Permission stage.
>
> **Structured context design:** For organisations with transversal teams (database, networking, OS) that need cross-application access, a flat context value (e.g., `"checkout"`) does not scale. Encoding `comp:<component>/bu:<business-unit>/app:<application>` into the context string, then using `MATCH('comp:db*')` in the IAM policy, enables precise transversal access without a per-application context per team.

> The dimension needed for transversal slicing should come first in the string for compatibility with Classic entity `startsWith` matching. See **IAM-04: Policy Authoring** for the complete design pattern and **IAM-05: Boundary Design** for boundary examples.
>
> **Critical:** When `dt.security_context` holds an array value, IAM policies must use the `MATCH` operator — not `=`, `STARTSWITH`, or `IN`. Array comparison with equality operators always returns false. See **ORGNZ-06** for the full rules.

```
// Check: Do your logs have security context assigned?
fetch logs, from:-1h
| summarize total = count(),
    with_context = countIf(isNotNull(dt.security_context)),
    without_context = countIf(isNull(dt.security_context))
| fieldsAdd coverage_pct = round(toDouble(with_context) / toDouble(total) *
100, decimals: 1)
```

```
// Check: Do your spans have security context assigned?
fetch spans, from:-1h
| summarize total = count(),
    with_context = countIf(isNotNull(dt.security_context)),
    without_context = countIf(isNull(dt.security_context))
| fieldsAdd coverage_pct = round(toDouble(with_context) / toDouble(total) *
100, decimals: 1)
```

7. Ingestion-Time vs. Query-Time Processing

A critical design decision: should you process data in OpenPipeline (at ingestion) or in DQL (at query time)? The answer depends on what you are trying to achieve.

When to Process at Ingestion (OpenPipeline)

Use Case	Why Ingestion Time
Drop noise	Data you never want to see should never be stored. Dropping debug logs, health check spans, or synthetic test events saves cost permanently.
Mask sensitive data	PII must be redacted before storage for compliance. You cannot "un-store" data.

Use Case	Why Ingestion Time
Add security context	`dt.security_context` must be set at ingestion — it cannot be added retroactively.
Extract metrics	Converting log patterns or span attributes into metrics happens at ingestion. You cannot create metrics from historical data.
Route to buckets	Bucket assignment is permanent at ingestion time. You cannot move data between buckets after storage.

When to Process at Query Time (DQL)

Use Case	Why Query Time
Exploratory analysis	When you do not yet know what you are looking for, keep the raw data and query flexibly.
Changing requirements	If the transformation logic changes frequently, DQL queries are easier to update than pipeline configurations.
Ad-hoc parsing	One-time investigations benefit from DQL `parse` rather than permanent pipeline processors.
Complex joins	Correlating across multiple data sources is a query-time operation.
Aggregation and visualization	Dashboard queries, trend analysis, and statistical aggregations belong in DQL.

A Third Surface: At-Source Enrichment on OneAgent (OneAgent 1.343)

The choice above is about *where inside Dynatrace* processing happens. OneAgent 1.343 adds a placement surface **upstream of both**: OneAgent ingests enrichment configuration directly and applies tags and metadata conditionally using **DQL matcher functions** (equality, phrase, and existence checks) at the point of capture — before the record reaches OpenPipeline.

Surface	Runs	Use it for
OneAgent enrichment (1.343+)	At the source, before ingest	Attributes only knowable on the host — process, container, or host-local metadata you would otherwise reconstruct downstream

Surface	Runs	Use it for
OpenPipeline	At ingestion	Everything in the first table above: drop, mask, security context, metric extraction, bucket routing
DQL	At query time	Everything in the second table above: exploration, ad-hoc parsing, joins, aggregation

Rollout is **per agent fleet, not per tenant** — **verify the OneAgent version on the hosts concerned**; the tenant version is not the agent version, and fleets commonly lag by more than one sprint. On OneAgent 1.342 and earlier the two-surface model above is the whole picture.

> **Enriching at the source does not displace OpenPipeline.** Security context, bucket routing, and masking remain OpenPipeline responsibilities — the gain is that a field is already present *when* OpenPipeline evaluates its matchers, so pipeline logic that would otherwise derive it can just match on it.

The Decision Framework

Ask these questions in order:

1. **Is the data sensitive?** → Mask at ingestion. No exceptions.
2. **Is the data noise?** → Drop at ingestion. Do not pay to store it.
3. **Does it need access control?** → Add security context at ingestion.
4. **Do you need a metric from it?** → Extract at ingestion.
5. **Everything else?** → Store raw, process at query time.

Summary

In this notebook you learned:

- **Configuration scopes** — OpenPipeline processes logs, spans, metrics, events, business events, security events, and more, each independently
- **Shared architecture** — All scopes follow the same flow (ingest → routing → pipeline → storage); inside the pipeline a fixed sequence of stages runs Processing (mask, drop,

transform) first and Bucket assignment before Metric extraction, Davis, and Data extraction

- **The default pipeline anti-pattern** — Sending everything through the default pipeline causes query performance, cost, security, and blast radius problems
- **Pipeline design principles** — One pipeline per source type, filter early, differentiate retention, order routing rules from specific to general
- **Matching conditions** — Conditional logic within a pipeline: every processor carries its own DQL matcher, so one pipeline can handle several data formats. Use matchers for different processing, separate pipelines for a different lifecycle or owner, and pipeline groups for centrally mandated stages.
- **Security context across scopes** — `dt.security_context` must be set at ingestion; data does not inherit an entity's security context, so extension metrics need it set in the extension configuration or the Permission stage
- **Ingestion vs. query time** — Process at ingestion for permanent actions (drop, mask, route, extract); process at query time for flexible analysis

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Next Steps

Continue to **OPIPE-02: Span Processing & Enrichment** to configure OpenPipeline for distributed traces — filtering noisy spans, enriching attributes, and routing traces to dedicated buckets. To run a pipeline group — ownership, mandated stages, execution order and gating — see **OPIPE-07: Pipeline Group Governance Patterns**.

References

- OpenPipeline (DT docs)
- What's new in Dynatrace SaaS 1.346 (DT docs) — the `dt.bindplane.*` namespace reservation quoted in §2
- Processing in OpenPipeline (DT docs)
- OpenPipeline data flow (DT docs)
- Metric extraction stage (DT docs)
- Davis stage (DT docs)
- Data extraction stage (DT docs)

- Primary Grail fields and tags (DT docs)
- OpenPipeline ingest sources (DT docs)
- OpenPipeline pipeline groups (DT docs)
- DQL matcher in OpenPipeline (DT docs) — `matchesValue`: *"Accepts wildcard * at the beginning or at the end of the value"*; the page documents `matchesPhrase`, `matchesValue`, `isNull` and `isNotNull`, and `contains()` / `in()` are rejected by the matcher verifier
- Use Grail buckets to partition data (DT docs)
- Security context for access control (DT docs)
- SaaS 1.344 release notes (DT docs)
- OneAgent 1.343 release notes (DT docs)

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